

A Design Study Approach to Classical Control

Randal W. Beard Timothy W. McLain Marc D. Killpack
Brigham Young University

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Homework D.2

- (a) Using the configuration variable z , write an expression for the kinetic energy of the system.
- (b) Create an animation of the mass-spring-damper system. The input should be a variable z . Turn in a screen capture of the animation.

Solution

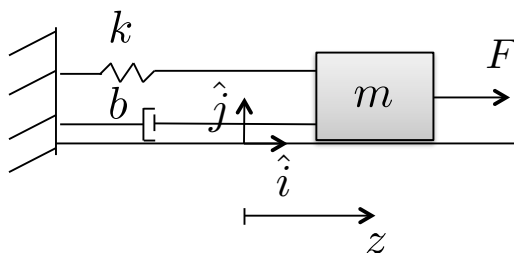


Figure 1: Computing the kinetic energy for the mass-spring-damper.

Define the inertial coordinate frame as in Figure 1, with $\hat{k}$ out of the page. The horizontal position of the mass m is given by

$$\mathbf{p} = \begin{pmatrix} z(t) \\ 0 \\ 0 \end{pmatrix}.$$

Differentiating to obtain the velocity of m

$$\mathbf{v} = \begin{pmatrix} \dot{z} \\ 0 \\ 0 \end{pmatrix}.$$

Since there is no rotational motion of the mass, the kinetic energy of the system is given by

$$\begin{aligned} K &= \frac{1}{2} m \mathbf{v}^\top \mathbf{v} \\ &= \frac{1}{2} m \begin{pmatrix} \dot{z} & 0 & 0 \end{pmatrix} \begin{pmatrix} \dot{z} \\ 0 \\ 0 \end{pmatrix} \\ &= \frac{1}{2} m \dot{z}^2. \end{aligned} \tag{1}$$